Comparative Study and Implementation of Generative AI Models

AE • VAE • DCGAN • Stable Diffusion

Assignment No. 2 | CO3 and CO4 | Fashion-MNIST experiments on Google Colab (Tesla T4)

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Reconstruction vs generation

Reconstruction

A model compresses a given image into a code and rebuilds it, so output can be compared with the input.

AE, VAE (CO3)

Generation

A model creates a new image from random noise or a text prompt; there is no single correct output.

DCGAN, Stable Diffusion (CO4)

Objectives

- Run AE and VAE reconstruction and compare blur and lost detail
- Train a compact DCGAN and inspect its samples
- Generate images with pretrained Stable Diffusion from text prompts
- Compare all outputs without overstating the results

Four models, four different jobs

Autoencoder

Reconstruction

How it works
Encoder compresses to a 64-value code; decoder rebuilds the image.

Strength / limit
Simple, fast. Blurs detail; not built for sampling.

VAE

Reconstruction + sampling

How it works
Encoder predicts mean and variance; loss = reconstruction + KL divergence.

Strength / limit
Smooth latent space. Blurrier output.

DCGAN

Generation from noise

How it works
Generator and discriminator trained against each other.

Strength / limit
Sharp edges. Unstable training, broken samples.

Stable Diffusion

Text-to-image

How it works
Pretrained latent diffusion removes noise step by step, guided by the prompt.

Strength / limit
Realistic and detailed. Heavy compute, prompt-sensitive.

AE, VAE and DCGAN were trained on Fashion-MNIST; Stable Diffusion is pretrained and used for inference only, so this compares capabilities, not a same-domain benchmark.

Setup and architectures

- Data: Fashion-MNIST, 60,000 train / 10,000 test, 28 × 28 grayscale, 10 classes
- Colab, PyTorch 2.11.0, Tesla T4 GPU; batch size 128; 10 epochs each; no seed set
- Stable Diffusion: runwayml/stable-diffusion-v1-5, float16, 20 steps, guidance 7.0, 512 × 512

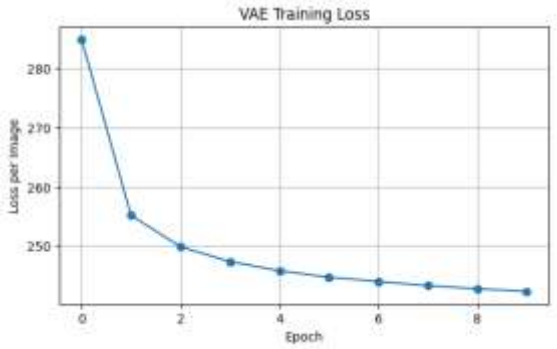
Model	Layers	Training	Parameters
Autoencoder	784→256→64 (latent)→256→784; ReLU, Sigmoid	Adam 1e-3, MSE	435,536
VAE	784→400; mean and log-variance (20 each); 20→400→784	Adam 1e-3, BCE + KL	652,824
DCGAN generator	Linear 100→6272; 2 transposed convs; BatchNorm, ReLU, Tanh	Adam 2e-4, BCE	778,305
DCGAN discriminator	2 convs + linear; LeakyReLU, Sigmoid	Adam 2e-4, BCE	138,817
Stable Diffusion	Pretrained text encoder, U-Net, decoder	None (inference)	Not counted

Training behaviour over 10 epochs

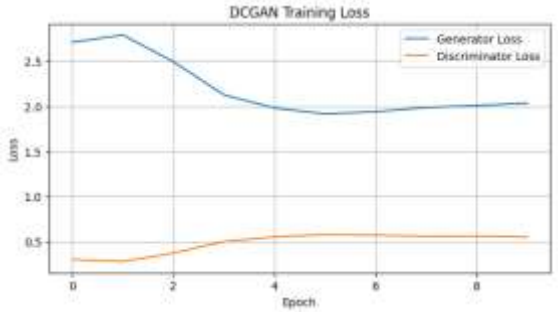
AE (MSE)



VAE (loss per image)



DCGAN (D and G loss)

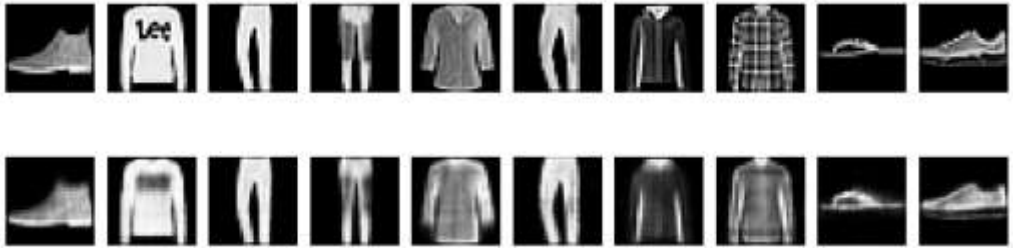


Model	Epoch 1	Epoch 10
AE loss (MSE)	0.033584	0.009718
VAE loss	284.9157	242.3201
DCGAN D / G loss	0.3026 / 2.7064	0.5507 / 2.0316

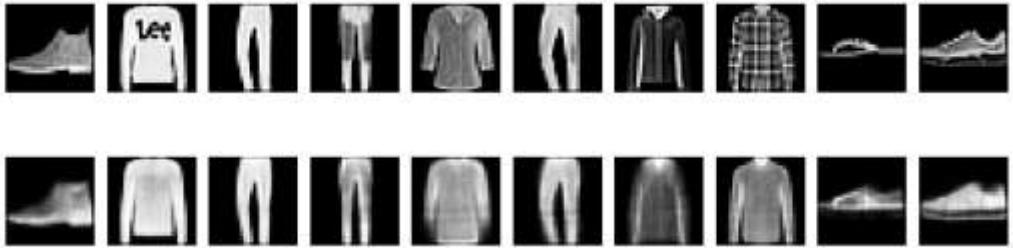
- AE and VAE losses fall fast, then slowly
- DCGAN losses level off from about epoch 5
- Losses use different scales; not comparable
- GAN losses are not image-quality scores

Reconstruction: the AE is more faithful than the VAE

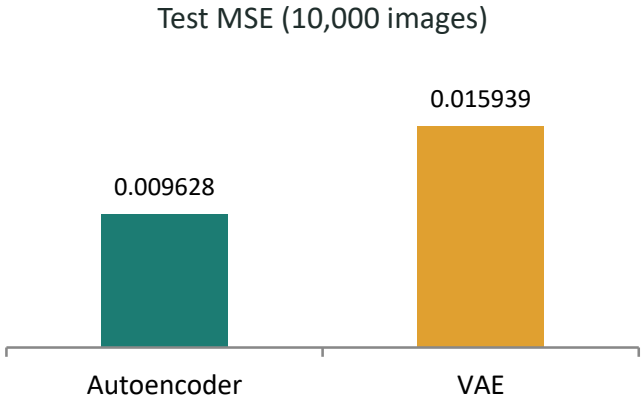
Autoencoder



VAE



Top row: original. Bottom row: reconstruction (first 10 test images).



- AE keeps logo and plaid faintly; VAE smooths them away
- VAE blur comes from the KL term and sampled latent vector
- Single run; models differ in size and loss

DCGAN: new images, but rough



64 samples generated from random noise after 10 epochs

What works

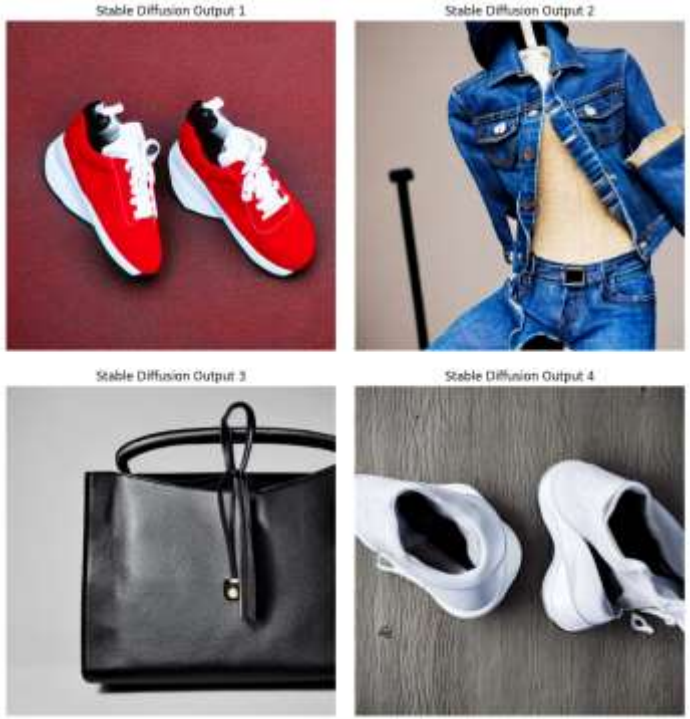
- Recognisable sneakers, boots, tops, T-shirts and trousers
- Sharper edges than the VAE reconstructions

What does not

- Many noisy, grainy or broken samples
- Several tops look alike (diversity not measured)
- Needs more training for clean results

Noise size 100 | Adam lr 2e-4 | final D / G loss 0.5507 / 2.0316

Stable Diffusion v1.5: realistic, with prompt misses



Red sneakers, denim jacket, leather handbag, white running shoes

#	Observation
1	Photo-like red sneakers, but on a red surface, not white
2	Clear denim stitching and buttons; odd mannequin and stray pole
3	Convincing leather grain and studio light; strap slightly tangled
4	Realistic mesh texture; shoes overlap, grey floor not white

20 steps | guidance 7.0 | 512 × 512 | safety checker disabled

No numerical quality score was recorded for DCGAN or Stable Diffusion.

Side-by-side comparison from our results

Criterion	AE	VAE	DCGAN	Stable Diffusion
How judged	Match to input (MSE 0.009628)	Match to input (MSE 0.015939)	Visual inspection	Realism and prompt match
Sharpness seen	Soft	Softest	Fairly sharp edges	Sharp and detailed
Creates new images	Not designed to	By sampling (not run here)	From noise	From text
Training effort	10 epochs	10 epochs	10 epochs, two networks	None (pretrained)
Control	Input image	Input or latent	Almost none	Text prompt
Main weakness	Blur	Stronger blur	Noisy, broken samples	Prompt misses, GPU need

Qualitative comparison based on this study; models use different data, so it is not a benchmark.

Conclusion

- No single best model: choose by goal (reconstruction, sampling or controllable generation)
- AE and VAE blur detail; the VAE blurred more (MSE 0.015939 vs 0.009628)
- DCGAN makes new images but needs more training for clean results
- Stable Diffusion gave the most realistic output, at higher compute cost and with prompt mismatches
- Limits: 10-epoch training, single run, different data across models

AI tools: ChatGPT and Claude. ChatGPT prompts covered Colab code for all four models, AE/VAE training and reconstruction plots, and debugging DCGAN, Stable Diffusion and MSE.
Verification: numbers and figures checked against the notebook output.
Nilesh: dataset, AE, VAE, MSE. **Bhusan:** DCGAN, Stable Diffusion, report and slides.